

SOLUTIONS ...

* For Glucose $i=1$ Always

SUB-TOPICS

Pg No. 30 * (2) Pys +
* From EAMCET

• General Concept & Formulas And Relation!

⇒ Molarity

⇒ Molality

⇒ Mole fraction

⇒ Normality

⇒ PPM

⇒ $\frac{W}{W}\%$, $\frac{V}{V}\%$, $\frac{W}{V}\%$ Percentages

• Henry's law

• Raoult's law

• Vapour Pressure and factors affecting it

• Raoult's law derivation by Dalton's law

• Volatile Non Volatile Substances

• Graphs of Positive Negative ideal Solutions!

• Table of Ideal, +ve, -ve!

• Examples of Positive deviations Negative deviations

• Fractional distillation and Azeotropes

• Colligative Properties

⇒ RLVP

⇒ EBP, DFP

⇒ OP

• Ostwald-Walker Process

• Van't Hoff factors!

• Type of Solutions!

• Molality = $M \times \frac{P}{1000}$ ($\because$ for Water $P=1000$)

FORMULAS

KEY CONCEPTS

$$\text{NO. OF MILLIMOLES} = M \times V_{(ml)}$$

$$\bullet \text{ Solution} = \text{Solvent}_{(Max)} + \text{Solute}_{(Min)}$$

Concentration terms:

- Molarity
- Molality
- Mole fraction
- Normality
- PPM

$$M_1 V_1 = M_2 V_2$$

$$N_1 V_1 = N_2 V_2$$

$$M_{\text{resultant}} = \frac{M_1 V_1 + M_2 V_2}{V_{\text{resultant}}}$$

$$\& N_{\text{resultant}} = \frac{N_1 V_1 + N_2 V_2}{V_{\text{resultant}}}$$

$$\ast \text{ PPM} = \frac{W}{W} \times 10^4$$

$$M = \frac{\text{wt Solute}}{\text{M} + \text{wt Solute}} \times \frac{1000}{V_{ml} \text{ Solution}}$$

$$m = \frac{\text{wt Solute}}{\text{Mw Solute}} \times \frac{1000}{\text{wt Solvent}}$$

$$P = P_B^0 x_B + P_A^0 x_A$$

Vapour phase:

$$y_B = \frac{x_B P_B^0}{P}$$

Relation B/w Molarity and molality:

$$m = \frac{M \times 1000}{1000 - M \times \text{wt Solute}}$$

Relation B/w Normality and Molarity:

$$M \times n\text{-factor} = \text{Normality}$$

$$N = \frac{\text{wt}_{\text{Solute}}}{\text{Mwt}_{\text{Solute}}} \times \frac{n \times 1000}{V_{\text{m}}^{\text{Solvent}}}$$

$$N = \frac{\text{NO. of Gram Equivalents}}{V_{\text{(L)}}} \quad (\because \text{NO. of Gram Eq.})$$

$$= \frac{\text{wt}}{\text{Eq. wt}}$$

Relation b/w Molarity, Molality, Mole fraction

$$\Rightarrow m = \frac{x_{\text{Solute}} \times 1000}{(1 - x_{\text{Solute}}) \times \text{Mwt}_{\text{Solvent}}}$$

$$\Rightarrow M = \frac{1000 d \times x_{\text{Solute}}}{M x_{\text{Solute}} + M x_{\text{Solvent}}}$$

Henry's Law: $P_A = K_H x_{\text{Solute}}$ ($\because \text{Temp} \propto \frac{1}{\text{Solubility}}$)

Raoult's Law:

$$P_T = x_A (P_A^\circ - P_B^\circ) + P_B^\circ$$

$$P_T = x_B (P_B^\circ - P_A^\circ) + P_A^\circ$$

$$\Rightarrow P_A^\circ = P_T x_A$$

$$\Rightarrow P_B^\circ = P_T x_B$$

Factors Affecting V.P:

$$\bullet \quad (VP)_{\text{Polar}} > (VP)_{\text{Non Polar}}$$

$$\bullet \quad VP \propto \text{Temp.}$$

$$\bullet \quad VP \propto \frac{1}{BP}$$

$$\bullet \quad VP \propto \frac{1}{\text{attractions}}$$

$$\bullet \quad VP \propto \frac{1}{\text{Mwt}}$$

$$\bullet \quad VP \text{ is independent of Surface area (container)}$$

$$\frac{Y_A}{Y_B} = \frac{P_A^\circ}{P_B^\circ} \times \frac{x_A}{x_B} \quad \frac{Y_A}{Y_B} = \frac{P_A^\circ}{P_B^\circ} \times \frac{n_A}{n_B}$$

$$Y_A = \frac{P_A^\circ x_A}{P_T}$$

A → Solvent & B → Solute

Lowering in VP is NOT a Colligative Property

$$** \frac{P^0 - P_s}{P_s} = \frac{n_B}{n_A} = \frac{x_B}{x_A} = \frac{m \times \text{Molecular wt of Solvent}}{1000} = \frac{w_B}{M_B} \times \frac{M_A}{w_A}$$

$$\frac{P^0 - P_{sol}}{P^0} : \text{Relative lowering VP.}$$

$$P^0 - P_{sol} : \text{Lower VP.} \quad \frac{P^0 - P_s}{P^0} = x_B$$

AZEOTROPIC MIX:

• $x = y$ distillation NOT Possible

• $x \neq y$ distillation Possible.

$$K_b = \frac{Mwt_A \times R \times T_b^2}{\Delta H_{vap} \times 1000}$$

$$K_f = \frac{Mwt_A \times R \times T_b^2}{\Delta H_b \times 1000}$$

$$\Rightarrow \Delta T_b = T_b - T_b^0$$

$$\Rightarrow \Delta T_f = T_f^0 - T_f$$

$$* \Delta T_f = K_f \times m$$

$$* \Delta T_b = K_b \times m$$

: 760 mm of Hg = 1 ATM.

$$i = \frac{\pi_{obs}}{\pi_{cal}}$$

$$i = 1 + (n-1)\alpha$$

$i = 1$ NO ASS DISS

$i < 1$ Association.

$i > 1$ Dissociation.

$$i = \frac{\text{Total NO. of Part Ass/Dis}}{\text{NO. of Part Taken}}$$

$$= \frac{\text{Exp/Obs/Abnormal}}{\text{Calculated/Theoretical}}$$

$$= \frac{\text{Normal Mass}}{\text{Abnormal Mass}}$$

Types of Solutions:

1. ISOTONIC $\pi_1 = \pi_2$

2. HYPERTONIC

$\pi_1 > \pi_2$ 1st solution
is hypertonic w.r.to 2nd

3. HYPOTONIC:

$\pi_1 < \pi_2$ 2nd soln

is hypotonic
w.r.to 1st.

Note: For Dil Solutions Molarity $\approx$ Molality //

Ideal

+ve

-ve

$$P_T = P_A + P_B$$

$$P_T > P_A + P_B$$

$$P_T < P_A + P_B$$

$$A-A = B-B = A-B$$

$$\frac{A-A}{B-B} > A-B$$

$$\frac{A-A}{B-B} < A-B$$

$$\Delta G = -ve$$

$$\Delta G = -ve$$

$$\Delta G = -ve$$

$$\Delta S = +ve$$

$$\Delta S = +ve$$

$$\Delta S = +ve$$

$$\Delta V_{\text{mix}} = 0$$

$$\Delta V_{\text{mix}} > 0$$

$$\Delta V_{\text{mix}} < 0$$

$$\Delta H = 0$$

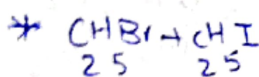
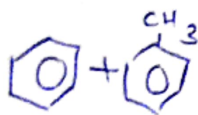
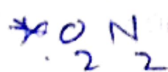
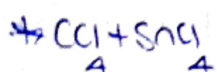
$$\Delta H > 0$$

$$\Delta H < 0$$

EXAMPLES:

* Diluted Solutions

* Benzene + Toluene

* n-heptane
n-hexane

* n-hexane n-heptane

* Ethyl bromide
Ethyl chloride

* Chlorobenzene bromobenzene

* Ethylene chloride Ethylene bromide

EXAMPLES:

* Water methanol
water Ethanol

* Acetone Alcohol

* $\text{CCl}_4 + \text{AlCl}_3$ * $\text{CCl}_4 + \text{Chloroform}$ * Water
+ Pyridine* Ethanol + CCl_4 * Ethyl acetate
and CS_2

EXAMPLES:

* Chloroform
+ acetone* acetone +
aniline* Water
+ ionisable
acid* $\text{HCl} + \text{water}$ * $\text{HNO}_3 + \text{H}_2\text{O}$ * Chloroform
Acetone* Chloroform
Benzene

Cross Check Hydrogen Bonding Such that

More interactions : negative deviation

less interactions : Positive deviation

* Type of Solutions :

1. Isotonic Solution :

Two Solutions having Same Osmotic Pressure

$$\pi_1 = \pi_2$$

2. Hypertonic :

$\pi_1 > \pi_2$ 1st Solution is hypertonic wto 2nd

3. Hypotonic :

$\pi_1 < \pi_2$ 2nd Solution is hypotonic wto 1st

When intermolecular Attraction b/w two components A and B in Mixture is Same as b/w A A or B B it is a case of ideal Solution.

When intermolecular Attraction b/w A B in A Mixture is Smaller than that b/w A A or B B Mixture is More vaporised BP is lowered it is case of Positive deviation from Raoult's law.

When intermolecular Attraction b/w A B is higher than that b/w A A or B B the Mixture is less vaporised BP is increased it is a case of negative deviation.

Methanol $\rightarrow$ CH_3OH

Weak & Strong Interactions:

Acetone $\rightarrow$ CH_3COCH_3 (Polar) 1) First check

Phenol $\rightarrow$ $\text{C}_6\text{H}_5\text{O}$ (Polar) H-bonding

Aniline $\rightarrow$ $\text{C}_6\text{H}_5\text{NH}_2$ 2) Dipole Dipole interactions

Toluene $\rightarrow$ CH_3 - C_6H_5 (Polar) 3) like, Unlike —
Attached to Ph

Chloroform $\rightarrow$ CCl_3 (Non Polar)

* * * Write Equation when dissociation involves calculations of i

